

CUSTOMERS TABLE:

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
CustomerID	FirstName	LastName	Email	RegistrationDate
1	John	Doe	john.doe@email.com	2022-03-15
2	Jane	Smith	jane.smith@email.com	2021-11-02
3	Alex	Taylor	alex.taylor@email.com	2023-01-10
4	Emily	Brown	emily.brown@email.com	2023-05-22
5	Michael	Davis	michael.davis@email.com	2023-08-14
* NULL	NULL	NULL	NULL	NULL

ORDERS TABLE:

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
OrderID	CustomerID	OrderDate	TotalAmount	
101	1	2023-07-01	150.50	
102	2	2023-07-03	200.75	
103	1	2023-08-10	450.00	
104	3	2023-08-12	89.99	
105	2	2023-09-01	310.20	
106	4	2023-09-15	75.50	
* NULL	NULL	NULL	NULL	

EMPLOYEES TABLE:

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmployeeID	FirstName	LastName	Department	HireDate	Salary
▶	1	Mark	Johnson	Sales	2020-01-15	50000.00
	2	Susan	Lee	HR	2021-03-20	55000.00
	3	Robert	Wilson	Sales	2019-06-11	62000.00
	4	Jessica	Garcia	IT	2022-02-01	75000.00
	5	David	Miller	IT	2021-08-15	71000.00
✱	NULL	NULL	NULL	NULL	NULL	NULL

QUERIES:

1. INNER JOIN

```
6 -- 1. INNER JOIN
7 • SELECT o.OrderID, o.OrderDate, o.TotalAmount, c.CustomerID, c.FirstName, c.LastName, c.Email
8 FROM Orders o
9 INNER JOIN Customers c ON o.CustomerID = c.CustomerID;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	OrderID	OrderDate	TotalAmount	CustomerID	FirstName	LastName	Email
▶	101	2023-07-01	150.50	1	John	Doe	john.doe@email.com
	103	2023-08-10	450.00	1	John	Doe	john.doe@email.com
	102	2023-07-03	200.75	2	Jane	Smith	jane.smith@email.com
	105	2023-09-01	310.20	2	Jane	Smith	jane.smith@email.com
	104	2023-08-12	89.99	3	Alex	Taylor	alex.taylor@email.com
	106	2023-09-15	75.50	4	Emily	Brown	emily.brown@email.com

2. LEFT JOIN

```
9      -- 2. LEFT JOIN
10 •    SELECT c.CustomerID, c.FirstName, c.LastName, c.Email, o.OrderID, o.OrderDate, o.TotalAmount
11      FROM Customers c
12      LEFT JOIN Orders o ON c.CustomerID = o.CustomerID;
```

Result Grid							
		Filter Rows:		Export:	Wrap Cell Content:		
	CustomerID	FirstName	LastName	Email	OrderID	OrderDate	TotalAmount
▶	1	John	Doe	john.doe@email.com	101	2023-07-01	150.50
	1	John	Doe	john.doe@email.com	103	2023-08-10	450.00
	2	Jane	Smith	jane.smith@email.com	102	2023-07-03	200.75
	2	Jane	Smith	jane.smith@email.com	105	2023-09-01	310.20
	3	Alex	Taylor	alex.taylor@email.com	104	2023-08-12	89.99
	4	Emily	Brown	emily.brown@email.com	106	2023-09-15	75.50
	5	Michael	Davis	michael.davis@email.com	NULL	NULL	NULL

3. RIGHT JOIN

```
14     -- 3. RIGHT JOIN
15 •    SELECT o.OrderID, o.OrderDate, o.TotalAmount, c.CustomerID, c.FirstName, c.LastName
16      FROM Orders o
17      RIGHT JOIN Customers c ON o.CustomerID = c.CustomerID;
```

Result Grid						
		Filter Rows:		Export:	Wrap Cell Content:	
	OrderID	OrderDate	TotalAmount	CustomerID	FirstName	LastName
▶	101	2023-07-01	150.50	1	John	Doe
	103	2023-08-10	450.00	1	John	Doe
	102	2023-07-03	200.75	2	Jane	Smith
	105	2023-09-01	310.20	2	Jane	Smith
	104	2023-08-12	89.99	3	Alex	Taylor
	106	2023-09-15	75.50	4	Emily	Brown
	NULL	NULL	NULL	5	Michael	Davis

4. FULL OUTER JOIN

```
19     -- 4. FULL OUTER JOIN
20 •    SELECT c.CustomerID, c.FirstName, o.OrderID, o.TotalAmount
21      FROM Customers c
22      LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
23      UNION
24      SELECT c.CustomerID, c.FirstName, o.OrderID, o.TotalAmount
25      FROM Customers c
26      RIGHT JOIN Orders o ON c.CustomerID = o.CustomerID;
```

Result Grid				
		Filter Rows:		Export:
	CustomerID	FirstName	OrderID	TotalAmount
▶	1	John	101	150.50
	1	John	103	450.00
	2	Jane	102	200.75
	2	Jane	105	310.20
	3	Alex	104	89.99
	4	Emily	106	75.50
	5	Michael	NULL	NULL

5. Subquery

```
28 -- 5. Subquery to find customers who have placed orders worth more than the average amount
29 • SELECT CustomerID, FirstName, LastName, Email FROM Customers WHERE CustomerID
30 IN (SELECT CustomerID FROM Orders WHERE TotalAmount > (SELECT AVG(TotalAmount) FROM Orders));
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
CustomerID	FirstName	LastName	Email	
1	John	Doe	john.doe@email.com	
2	Jane	Smith	jane.smith@email.com	
NULL	NULL	NULL	NULL	

6. Subquery

```
32 -- 6. Subquery to find employees with salaries above the average salary
33 • SELECT EmployeeID, FirstName, LastName, Department, Salary FROM Employees
34 WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
EmployeeID	FirstName	LastName	Department	Salary
4	Jessica	Garcia	IT	75000.00
5	David	Miller	IT	71000.00
NULL	NULL	NULL	NULL	NULL

7. Extract

```
36 -- 7. Extract the year and month from the OrderDate
37 • SELECT OrderID, OrderDate,
38 EXTRACT(YEAR FROM OrderDate) AS OrderYear,
39 EXTRACT(MONTH FROM OrderDate) AS OrderMonth
40 FROM Orders;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
OrderID	OrderDate	OrderYear	OrderMonth
101	2023-07-01	2023	7
102	2023-07-03	2023	7
103	2023-08-10	2023	8
104	2023-08-12	2023	8
105	2023-09-01	2023	9
106	2023-09-15	2023	9

8. Difference

```
42 -- 8. Calculate the difference in days between two dates (order date and current date)
43 • SELECT OrderID, OrderDate,
44 CURRENT_DATE AS TodayDate,
45 DATEDIFF(CURRENT_DATE, OrderDate) AS DaysDifference
46 FROM Orders;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
OrderID	OrderDate	TodayDate	DaysDifference
101	2023-07-01	2026-07-26	1121
102	2023-07-03	2026-07-26	1119
103	2023-08-10	2026-07-26	1081
104	2023-08-12	2026-07-26	1079
105	2023-09-01	2026-07-26	1059
106	2023-09-15	2026-07-26	1045

9. Format

```
48 -- 9. Format the OrderDate to a more readable format (e.g., 'DD-MMM-YYYY')
49 • SELECT OrderID, OrderDate,
50    DATE_FORMAT(OrderDate, '%d-%m-%Y') AS FormattedDate
51 FROM Orders;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
OrderID	OrderDate	FormattedDate	
101	2023-07-01	01-07-2023	
102	2023-07-03	03-07-2023	
103	2023-08-10	10-08-2023	
104	2023-08-12	12-08-2023	
105	2023-09-01	01-09-2023	
106	2023-09-15	15-09-2023	

10. Concatenate

```
53 -- 10. Concatenate FirstName and LastName to form a full name
54 • SELECT CustomerID, CONCAT(FirstName, " ", LastName) AS FullName FROM Customers;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
CustomerID	FullName		
1	John Doe		
2	Jane Smith		
3	Alex Taylor		
4	Emily Brown		
5	Michael Davis		

11. Replace

```
59 -- 11. Replace part of a string (e.g., replace 'John' with 'Jonathan')
60 • SELECT CustomerID, FirstName,
61    REPLACE(FirstName, 'John', 'Jonathan') AS UpdatedFirstName
62 FROM Customers;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
CustomerID	FirstName	UpdatedFirstName	
1	John	Jonathan	
2	Jane	Jane	
3	Alex	Alex	
4	Emily	Emily	
5	Michael	Michael	

12. Convert into upper and lower case

```
64 -- 12. Convert FirstName to uppercase and LastName to lowercase
65 • SELECT CustomerID,
66    UPPER(FirstName) AS UpperFirstName,
67    LOWER(LastName) AS LowerLastName
68 FROM Customers;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
CustomerID	UpperFirstName	LowerLastName	
1	JOHN	doe	
2	JANE	smith	
3	ALEX	taylor	
4	EMILY	brown	
5	MICHAEL	davis	

13. Trim extra spaces

```
70 -- 13. Trim extra spaces from the Email field
71 • SELECT CustomerID,
72     TRIM(Email) AS TrimmedEmail
73     FROM Customers;
74
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
CustomerID	TrimmedEmail			
3	alex.taylor@email.com			
4	emily.brown@email.com			
2	jane.smith@email.com			
1	john.doe@email.com			
5	michael.davis@email.com			

14. Running total

```
75 -- 14. Calculate the running total of TotalAmount for each order
76 • SELECT OrderID, CustomerID, OrderDate, TotalAmount,
77     SUM(TotalAmount)
78     OVER (ORDER BY OrderDate, OrderID) AS RunningTotal
79     FROM Orders;
80
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
OrderID	CustomerID	OrderDate	TotalAmount	RunningTotal
101	1	2023-07-01	150.50	150.50
102	2	2023-07-03	200.75	351.25
103	1	2023-08-10	450.00	801.25
104	3	2023-08-12	89.99	891.24
105	2	2023-09-01	310.20	1201.44
106	4	2023-09-15	75.50	1276.94

15. Rank orders

```
81 -- 15. Rank orders based on TotalAmount using the RANK() function
82 • SELECT OrderID, CustomerID, TotalAmount,
83     RANK() OVER (ORDER BY TotalAmount DESC) AS OrderRank
84     FROM Orders;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
OrderID	CustomerID	TotalAmount	OrderRank	
103	1	450.00	1	
105	2	310.20	2	
102	2	200.75	3	
101	1	150.50	4	
104	3	89.99	5	
106	4	75.50	6	

16. Assign discount

```
86 -- 16. Assign a discount based on TotalAmount in orders (> 1000: 10% off, > 500: 5% off)
87 • SELECT OrderID, TotalAmount,
88     CASE
89         WHEN TotalAmount > 1000 THEN '10% Off'
90         WHEN TotalAmount > 500 THEN '5% Off'
91         ELSE 'No Discount'
92     END AS Discount
93 FROM Orders;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
OrderID	TotalAmount	Discount	
101	150.50	No Discount	
102	200.75	No Discount	
103	450.00	No Discount	
104	89.99	No Discount	
105	310.20	No Discount	
106	75.50	No Discount	

17. Categorize salaries

```
95 -- 17. Categorize employees' salaries as high, medium, or low
96 • SELECT EmployeeID, FirstName, LastName, Salary,
97     CASE
98         WHEN Salary >= 70000 THEN 'High'
99         WHEN Salary >= 50000 THEN 'Medium'
100        ELSE 'Low'
101    END AS SalaryCategory
102 FROM Employees;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	EmployeeID	FirstName	LastName	Salary	SalaryCategory
1	Mark	Johnson	50000.00	Medium	
2	Susan	Lee	55000.00	Medium	
3	Robert	Wilson	62000.00	Medium	
4	Jessica	Garcia	75000.00	High	
5	David	Miller	71000.00	High	